Rank augmentation: complete manuscript and program costs

Human Compression project

The original is De Saedeleer–Leemans–Mulpas, *A rank augmentation theorem for rank 3 string C-group representations of the symmetric groups*, arXiv:2212.12563v1, 23 December 2022 (L020). All nineteen pages are counted, including the banner, introductory table, diagrams, references and addresses. The nineteen accepted dispositions preserve the valid first two cases of Theorem 4.1 and replace its false third case by an exact criterion and a complete eighteen-point counterexample. The original retains its literal claims in every measured representation. This is one original, not two.

The complete written comparison. For raw, untrimmed extraction in NFC, counting characters without whitespace,

$$R_{\text{written}} = \frac{33604}{13904 + (7242 + 6854)} = \frac{33604}{28000} = 1.200143.$$

The whole five-page common paper and both three-page local papers are paid. The counterproof is a required application dependency; citing an earlier correction does not make it free. No old Topic B common theory or cross-topic saving is assumed. Titles, references and repeated exposition remain charged.

Complete document	Pages	Characters
Original L020	19	33604
Common supported-transposition proof	5	13904
Source application	3	7242
Required counterexample proof	3	6854
All three programs, printed	3	10866

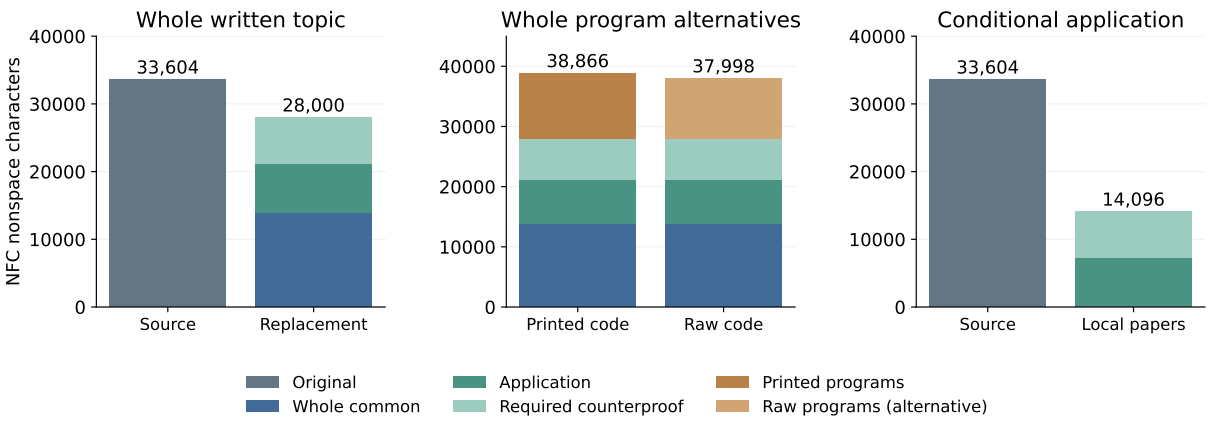
Complete programs and the conditional application. Including the whole program appendix gives the separate comparison $33604/(28000 + 10866) = 0.864612$. The three whole raw programs cost 9998 characters and give $33604/(28000 + 9998) = 0.884362$. Printed and raw programs are alternatives; the old two-page appendix is replaced, not added. The program-inclusive denominators exceed the source length in this coordinate.

After acquiring *the entire common paper, all three declared programs and all retained premises*, the complete six-page local application costs 14096 characters:

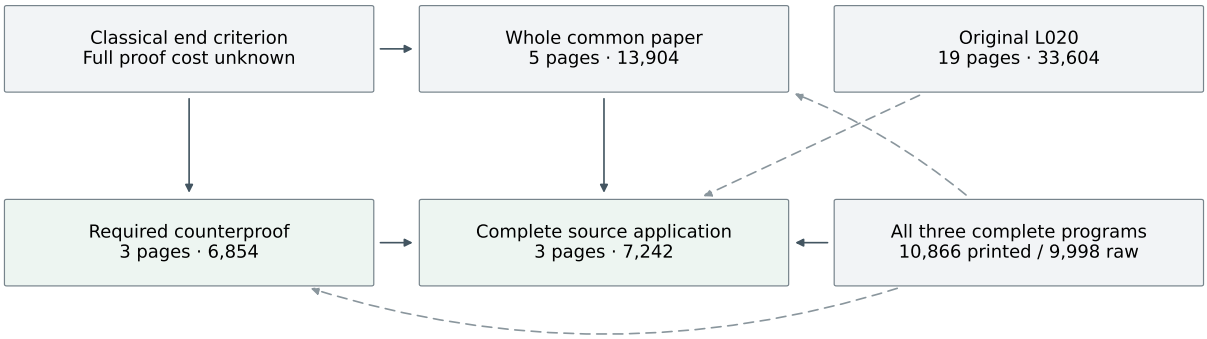
$$r \mid (U, \text{programs}) = 33604/(7242 + 6854) = 2.383939.$$

This is a conditional acquisition comparison, not the whole-topic factor. The source-example program supplies finite certificates; the model program is bounded validation; the eighteen-point certificate supports the complete written counterproof. No bounded test replaces a uniform proof.

Separate and unknown costs. Three drivers cost 14096 nonspace characters (15,947 literal bytes). Two whole Lean files cost 5218 characters (6,855 bytes), with two generic and thirteen literal-obstruction declarations, not a full source formalization. The motivating Table 1 census and classical end criterion remain retained; full outside-proof, runtime, historical-software, graphic and formalization costs are unknown, not zero. These are complete-text representation lengths, not minimum proof complexity or research effort. No new scientific credit is assigned by this numerical packet.



Conditional comparison prepays whole common paper, all three declared programs and retained premises.



Solid: proof / finite input → use. Dashed: source correspondence or validation.

The graph distinguishes proof dependencies from source correspondence and validation. It does not remove any document from the cost. In particular, the counterproof stays in the local application and the common paper is charged once. Whole-document acquisition sums and every raw file are also available in JSON/CSV; no node-count factor is used.

Unit and encoding sensitivities. Raw, untrimmed NFC coordinates are:

Unit	Written	Printed	Raw	Conditional
Whitespace tokens	1.365	1.202	1.226	2.817
Alphanumeric units	1.385	1.010	1.028	2.812
Nonspace characters	1.200	0.865	0.884	2.384
Collapsed-space characters	1.231	0.917	0.938	2.462
Collapsed-space UTF-8 bytes	1.158	0.886	0.904	2.361

All thirty coordinates and 120 exact fractions are retained: five units, NFC and NFKC, with untrimmed, furniture-selected and calibrated-selected text. Selection removes only exact repeated headers and isolated page numbers. Two actual font/byte classes account for 112 observed substitutions: 111 calligraphic G symbols and one blackboard-bold N. The raw coordinates are unchanged. The arXiv banner remains included; printed hyphens remain literal. Full diagram topology, colour, curvature and layout are not reconstructed from extracted labels, so graphic complexity stays unknown. Normalized text-byte units differ from literal stored-file bytes.

Review boundary. Scientific acceptance and the distinct input review are reused. The numerical author previously reviewed those inputs but did not author them; this new arithmetic and reader require a distinct numerical review. The source's larger-orbit and higher-rank questions remain unresolved. No source, question or formal credit is added here.